

Exercise 6: NULL FUNCTIONS

Q1: Show all employees with their salary. If salary is null, display 0.

⇒ employee - id, name, salary - with - default

Table 1: Employees

employee - id	name	department	salary
1	Alice	HR	500
2	Bob	IT	NULL
3	Charlie	NULL	7000
4	Dana	Finance	NULL

SELECT employee - id, name IFNULL (value, default) replaces null with the default 0.
IFNULL (salary, 0) AS salary - with - default

FROM employees;

employee - id	name	salary - with - default
1	Alice	500
2	Bob	0
3	Charlie	7000
4	Dana	0

Q2: Show employee names with their department. If department is null, show "Not assigned"

⇒ employee - id, name, department - name.

SELECT employee - id, name
IFNULL (department, 'Not assigned') AS department - name
FROM employees;

employee-id	name	department-name
1	Alice	HR
2	Bob	IT
3	Charlie	Not assigned
4	Dana	Finance

Q3 Find orders with null customer-id using ISNULL()

→ order-id, customer-id ⇒ ISNULL identifies rows with missing value

Table2: Orders

order-id	customer-id	delivery-date
101	201	2024-12-01
102	202	NULL
103	NULL	2024-12-03

SELECT order-id, customer-id

FROM orders

WHERE customer-id ISNULL;

order-id	customer-id
103	NULL

Q4. Show all orders. If delivery-date is null, show 'Pending'

→ order-id, customer-id, delivery status

SELECT order-id, customer-id

IFNULL (delivery-date, 'Pending') AS delivery status
FROM orders;

order-id	customer-id	delivery-status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

Q5 Show all students and their grades. Replace null with 0.
 ⇒ student-id, name, final-grade.

Table 3: Students

student-id	name	grade
1	Ethan	85
2	Maya	NULL
3	Olivia	90

SELECT student-id, name
 IF NULL (grade, 0) AS final-grade
 FROM students;

student-id	name	final-grade
1	Ethan	85
2	Maya	0
3	Olivia	90

Q6 Count students who haven't been graded.
 ⇒ not-graded-count.

SELECT COUNT(*) AS not-graded-count
 FROM students
 WHERE grade IS NULL;

not-graded-count
1

Q7. Show product name, price and final price after discount (assume 0 if discount is null)

⇒ product-id, name, final price

Table A: Products

#CONSULT!

product-id	name	price	discount
501	Keyboard	25	NULL
502	Mouse	15	5
503	Monitor	100	NULL

SELECT product-id, name

FROM Products;

product-id	name	final-price
501	Keyboard	
502	Mouse	
503	Monitor	

Q8. Count how many customers have no email.

⇒ missing-email-count

Table S: Customers

customers-id	name	email
1	Linda	NULL
2	Joseph	joseph@mail.com
3	Mia	NULL

SELECT COUNT(*) AS missing-email-count

FROM customers

WHERE email IS NULL;

Missing email - count
2

Q9. Show all customers with email. IF NULL, display "No email".

⇒ customer-id, name, email-display.

SELECT customer-id, name
IF NULL (email, 'No email') AS email-display.
FROM customers;

customer-id	name	email-display
1	Linda	No email
2	Joseph	joseph@email.com
3	Nid	No email

Q10. Show payment details with method replaced by "Unknown" if NULL.

⇒ payment-id, method-display, status

Table 6: Payments

payment-id	method	status
301	Credit	NULL
302	Payroll	Success
303	NULL	Failed

Select payment-id, status

IF NULL (method, 'Unknown') AS method-display

FROM payments;

payment-id	method-display	- status
301	Credit	Null
302	Paypal	Success
303	Unknown	Failed

Q11: Show items and their quantity (0 if null)

⇒ item-id, item-name, quantity-checked

Table 7: Inventory

item-id	item-name	quantity
1	Pen	Null
2	Notebook	150
3	Eraser	Null

SELECT items-id, item-name

IFNULL (quantity, 0) AS quantity-checked

FROM Inventory;

items-id	item-name	quantity-checked
1	Pen	0
2	Notebook	150
3	Eraser	0

Q12: Show employee-id and the first available value among bonus or commission.

⇒ emp-id, first-available-reward

Table 8: Employee-extra

emp-id	bonus	commission
1	Null	300
2	100	Null
3	Null	Null

COALESCE() returns the first non-null value!

SELECT emp-id

COALESCE (bonus, commission,) AS first-avail-reward

FROM employees-extra;

emp-id	first-available-reward
1	300
2	100
3	NULL

Q13. Count classes that don't have a room assigned

⇒ no-room-count

Table 9: Classes

class-id	subject	room
11	Math	NULL
12	Science	lab A
13	English	NULL

SELECT COUNT (*) AS no-room-count

FROM classes

WHERE room IS NULL;

no-room-count
2

Q14. Show attendance records with status. Replace null with

"Not Mailed"

⇒ student-id, date, attendance-status.

Table 10: Attendance

student-id	date	status
1	2025-04-01	NULL
2	2025-04-01	Present
3	2025-04-01	Absent

SELECT student-id, date

IFNULL (status, 'Not Marked') AS attendance-status
FROM Attendance;

student-id	date	Attendance-status
1	2025-04-01	Not marked
2	2025-04-01	Present
3	2025-04-01	Absent

Q15. Show account id, account type (or 'Unknown') and balance (or 0).

⇒ account-id, type-display, balanced-checked

Table 11: Bank-Accounts

account-id	account-type	Balance
A1	Savings	NULL
A2	Current	5000
A3	NULL	2000

SELECT account-id, type-display

IFNULL (account-type, 'Unknown') AS type-display,

IFNULL (balance, 0) AS balanced-checked

FROM Bank-Accounts;

account - id	type - display	balance checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

Q16. Show all projects with a start date. If start-date is Null, display 'TBD'.

⇒ project-id, title, start-display

Table 12: Projects

project-id	title	start-date	end-date
1	Website Revamp	2025-01-10	NULL
2	Mobile App	NULL	2025-06-01
3	Data Migration	NULL	NULL

SELECT project-id, title

IFNULL (start-date, 'TBD') AS start-display.

FROM Projects;

project-id	title	start-display
1	Website Revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

Q17. Display reviews showing comment (or 'No comment') and rating (or 0).

⇒ review-id, product-id, comment-display, rating-display

Table 13: Reviews

review-id	product-id	comment	rating
1	501	Great Product	4
2	502	NULL	NULL
3	503	Works fine	3

SELECT review-id, product-id

IFNULL (comment, 'No comment') AS comment-display

IFNULL (rating, 0) AS rating-display

FROM Reviews;

review-id	product-id	comment-display	rating-display
1	501	Great Product	4
2	502	No comment	0
3	503	Works fine	3

Q18. Show the supplier contact number. Use COALESCE (phone, all-phone, 'No contact')

⇒ supplier-id, name, contact-number

Table 14: Suppliers

supplier-id	name	phone	all-phone
1	Global Goods	NULL	123456789
2	Best Suppliers	987654321	NULL
3	Value Source	NULL	NULL

SELECT supplier-id, name

COALESCE (phone, all-phone, 'No contact') AS contact-number

FROM Suppliers;

supplier-id	name	Contact-number
1	Global Goods	123456789
2	Best Suppliers	987654321
3	Value Source	No contact / NULL

Q19. Show all users and their preferences. Replace all NULL with defaults: $\Rightarrow$ Theme $\rightarrow$ "Light"

$\Rightarrow$ language $\rightarrow$ "English"

$\Rightarrow$ Timezone $\rightarrow$ "UTC"

$\Rightarrow$ user-id, theme-set, language-set, timezone-set

Table 15: User-Settings

user-id	theme	language	timezone
1	NULL	ENGLISH	NULL
2	DARK	NULL	UTC+1
3	NULL	NULL	NULL

SELECT user-id

IFNULL (theme , 'light') AS theme-set,

IFNULL (language , 'English') AS language-set,

IFNULL (Timezone , 'UTC') AS timezone-set

FROM User-Settings.

user-id	theme-set	language-set	timezone-set
1	light	English	UTC
2	Dark	English	UTC+1
3	light	English	UTC

Q. Show maintenance log with:

→ issue → default to "Unknown Issue"

→ technician → default to "Not Assigned"

→ record-id, machine-id, issue-log, technician-name

Table 16: Maintenance

record-id	machine-id	issue	technician
1	M101	Overheating	NULL
2	M102	NULL	NULL
3	M103	Jammed	Alex

SELECT record-id, machine-id

IFNULL(issue, 'Unknown Issue') AS issue-log,

IFNULL(technician, 'Not Assigned') AS technician-name

record-id	machine-id	issue-log	technician-name
1	M101	Overheating	Not Assigned
2	M102	Unknown Issue	Not Assigned
3	M103	Jammed	Alex